



Autonomous System Deduplication for Monero Node Peer Selection

Draft v0.1

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Monero Project contributor. Method, simulation code and game-theoretic model adapted from Rucknium (Monero Research Lab)

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Abstract

Monero nodes deduplicate outbound peer candidates by /24 subnet (pull request #9939), following Rucknium [Rucknium, 2025a]. Pull request #11474 adds an opt-in `-asmmap` option that deduplicates by autonomous system (AS) instead. This note applies Rucknium's simulation and game to AS deduplication. In October 2026 the suspected spy nodes occupy 1,132 /24 subnets, but 98.9 percent of them are in one AS. Against them, AS deduplication reduces an honest node's outbound connections to suspected spy nodes from 4.35 to 0.13 out of 12 in simulation, and from 33.8 to 2.7 percent on live nodes. Honest nodes, however, occupy 3.46 times more /24 subnets than ASes, so an adversary that spreads over many ASes does better against AS deduplication. Servers in other ASes cost about the same as bulk servers, so the number of rentable ASes decides. AS deduplication stays better while the adversary can rent servers in fewer than about 370 ASes, and a sample survey estimates 475 (330 to 657) rentable ASes among those that already host Tor, Bitcoin or Monero servers. The results support AS deduplication as an opt-in option, not as the default. Choosing Dandelion++ stems uniformly over the ASes of the outbound connections, without changing peer selection, reduces the probability that a stem is a suspected spy node from 36.5 to 12.3 percent, and costs at most 3.3 percentage points against every adversary placement simulated.

1 Statement of the problem

Spy nodes can reduce the effectiveness of Dandelion++ [Fanti et al., 2018]. Rucknium showed that an adversary can exploit bulk prices for whole subnets, and that deduplicating outbound peer candidates by /24 subnet reduces the share of an honest node's outbound connections made to suspected spy nodes from 30.8 to 8.8 percent [Rucknium, 2025a]. The deduplication was implemented in Monero pull request #9939 [rbrunner7, 2025].

Subnet deduplication assumes that the adversary's nodes are packed into few subnets. An adversary that rents virtual servers from a large cloud provider gets addresses scattered over many /24 subnets, but all of them inside the provider's autonomous system (AS). An AS is a network under a single routing policy, identified by an AS number (ASN). Bitcoin Core has grouped peers by ASN, using a map from IP prefixes to ASNs called an *asmmap*, since 2020 [naumenkogs, 2019]. Monero pull request #11474 [plowsof, 2026b], a revival of #7935 [Malinero, 2021], adds an opt-in `-asmmap` option that groups peers by ASN for outbound peer selection.

When this was discussed on 5 October 2026, Rucknium accepted AS grouping as an opt-in option and stated the conditions for making it the default [#no-wallet-left-behind, 2026]: an analysis showing “with reasonable confidence, that the cure would not be worse than the disease”, as was done for /24 deduplication. He raised four concerns, which this note addresses in turn:

1. An adversary could spread across many ASNs in response to a default rollout, making the problem worse.
2. Many honest nodes are concentrated in the ASNs of server providers, so an adversary could exploit AS avoidance to obtain more outbound connections than honest nodes.
3. ASNs are harder to analyze than subnets because ASNs do not have a defined size.
4. Nodes in sparsely populated ASNs would receive many more inbound connections.

2 Background

2.1 Peer selection with /24 and AS deduplication

The selection rule is implemented in `make_new_connection_from_peerlist()` in `src/p2p/net_node.inl` [plowsof, 2026b]. For each new outbound connection in the public zone, a node:

1. collects the *groups* of its current connections. Without `-asmap` the group of an IPv4 address is its /24 subnet. With `-asmap` the group is the ASN that the asmap returns, and only outbound connections are counted;
2. shuffles its peer list and keeps the first candidate of each group that is not already a connected group;
3. picks one of the remaining candidates.

After steps 1 and 2 there is one candidate per group, so the probability of selecting a group does not depend on how many addresses the group contains. Appendix A shows that, when the final pick is uniform, the draw is equivalent to choosing a non-connected group uniformly and then a member of that group uniformly. This is why the size of an AS, concern 3 above, does not enter the analysis of an AS that contains only adversary nodes: a large AS and an AS with a single node are each one group.

Bitcoin Core connects out to at most one peer per network group, which is the ASN when an asmap is used, and also uses the group to place addresses in its address manager buckets [Bitcoin Core, 2026b, naumenkogs, 2019]; candidates in an already connected group are skipped (`src/net.cpp`, line 2915, same commit). It does not deduplicate candidates in the way described above. Its asmap was introduced against AS-level routing attacks and eclipse attacks [Apostolaki et al., 2017, Tran et al., 2020, Heilman et al., 2015], not against spy nodes. It is opt-in in Bitcoin Core: `-asmap` with no file uses an embedded map and `-asmap=<file>` reads a map from disk [Bitcoin Core, 2026b, fjahr, 2023]. Eclipse attacks on Monero’s peer-to-peer network are analyzed by [Shi et al., 2025].

2.2 The asmap

Pull request #11474 embeds `2026/1788801420_asmap.dat` from the `bitcoin-core/asmap-data` repository [bitcoin-core/asmap-data, 2026b]. It is a “filled” map: in this note’s own lookup of the map, every one of the 2^{32} IPv4 addresses resolves to an ASN, and 82,443 distinct ASNs appear in the IPv4 part of the map (replication code: `asmap_coverage.py`). The fallback to /24 groups for unmapped addresses in pull request #11474 is therefore never used for IPv4, and an adversary cannot use unmapped address space to get one group per /24.

3 Data

3.1 Box: Data and suspected spy node classification

October 2026 data. Reachable Monero nodes were collected by a crawler running on four vantage points [plowsof, 2026a]. A node is *reachable* if a handshake to its advertised address succeeded. The snapshot was taken on 2026-10-05 at 16:19 UTC and contains 3,983 distinct reachable IPv4 addresses.

A node was classified as a suspected spy node if its handshake showed one of two protocol fingerprints that an unmodified `monerod` does not produce. The fingerprints are withheld from this public draft so that the operators cannot evade them, and are available to the Monero Research Lab on request. As a check for false positives, the classifier was applied to the default remote nodes in the Monero GUI pull request #4601 [selsta, 2026]: of 157 hosts, 145 IPv4 addresses resolved, 80 of them were in the reachable dataset, and 0 of those were classified as suspected spy nodes.

Ban-listed nodes. The crawler does not handshake addresses in the MRL ban list [Rucknium, 2024], version 2, so the snapshot contains none of them. To measure them, every ban-listed IPv4 address that peers gossiped to the crawler, 2,440 in total, was handshaked from the four vantage points on 2026-10-05 (replication code: `ban_crawl.py`). 2,025 (83.0 percent) completed a handshake. They occupy 68 /24 subnets and 3 ASNs, 1,965 of them in AS401476 (SCNL-1-0001 [IPtoASN, 2026]). They are added to the October 2026 data in Section 7 to represent a node that does not use the ban list, which is the default.

May 2025 data. To make the results reproducible without the withheld classification, the network simulations and the price-premium response are repeated on the data used in [Rucknium, 2025a]: the `good_peers` network scan and the `ban_list_v1` ban list from the `xmrpeers` R package [Rucknium, 2025b], cleaned with the same code. This gives 4,607 reachable IPv4 addresses, of which 1,843 are in the ban list, the same totals as in [Rucknium, 2025a].

ASNs. All addresses in both datasets are mapped to ASNs with the `asmap` embedded by pull request #11474. The May 2025 addresses are therefore mapped with a map from 2026.

Table 1 summarizes how the nodes in each dataset are grouped. In October 2026, the 1,293 suspected spy nodes occupy 1,132 distinct /24 subnets but only 10 ASNs; 1,279 of them (98.9 percent) are in AS14061, which also hosts 630 honest reachable nodes. /24 deduplication leaves almost one group per suspected spy node, while AS deduplication reduces them to a few groups. Honest nodes are less concentrated: 2,690 honest nodes occupy 1,983 /24 subnets and 573 ASNs.

What the label means. Both fingerprints show that a node runs modified software. Neither shows what its operator does with the connections it receives, so “suspected spy node” is an inference and should be read as “node of a large fleet running modified software”. AS14061 is DigitalOcean (DIGITALOCEAN-ASN [IPtoASN, 2026]). The fleet is 67 percent of the reachable Monero nodes in that AS, not a large part of DigitalOcean; the AS that hosts a node is not necessarily its operator. The fleet is uniform. Of its 1,279 nodes in AS14061, 100 percent listen on the default port, 99.7 percent are within 10 blocks of the chain tip (honest nodes: 97.1), 99.8 percent are unpruned (honest: 67.0) and 99.7 percent advertise an RPC port (honest: 59.4), and every node has its own peer ID (replication code: `fleet_profile.py`). This is consistent with a single deployment, but the distinct peer IDs mean that a single operator cannot be shown from the handshake. [Kopyciok et al., 2026] describe the same pattern, many anomalous peers concentrated in one AS that promotes its own addresses, for June 2025, and argue that a consistent deviation from the network baseline is significant for security “regardless of ultimate intent”. That AS, LIONLINK-NETWORKS (AS54098), is also the largest AS among the ban-listed nodes in the May 2025 data (566 nodes), and it announces the address ranges of the entity that b10c calls LinkingLion in the Bitcoin network [b10c, 2024]. Benign explanations for the October 2026 fleet, such as a commercial remote-node service or a measurement project, cannot be excluded. None of the results in this note depends on intent: they measure how many of an honest node’s outbound connections and Dandelion++ stems one concentrated fleet obtains, which bounds what it could learn whether or not it records transactions [Fanti et al., 2018, Sharma et al., 2023].

Table 1: Grouping of reachable nodes by /16 subnet, /24 subnet, and autonomous system

Dataset	Nodes	IP addresses	/16 subnets	/24 subnets	ASNs	Effective ASNs
Oct 2026	Suspected spy	1,293	77	1,132	10	1.0
Oct 2026	Honest	2,690	1,273	1,983	573	13.2
May 2025	Honest	2,764	1,903	2,647	794	61.9
May 2025	Suspected spy	1,843	61	257	7	5.5

Figure 1 plots every reachable node in October 2026, grouped by ASN (black perimeters) and /24 subnet (yellow perimeters), in the style of [Rucknium, 2025a]. Figure 2 shows the groups that remain after deduplication, coloured by the share of suspected spy nodes in each group.

Figure 1: ASN treemap of honest and suspected spy nodes, October 2026

ASN treemap of honest and suspected spy nodes

Black perimeters indicate autonomous systems. Yellow indicates /24 subnets.

Node type: ■ honest ■ spy

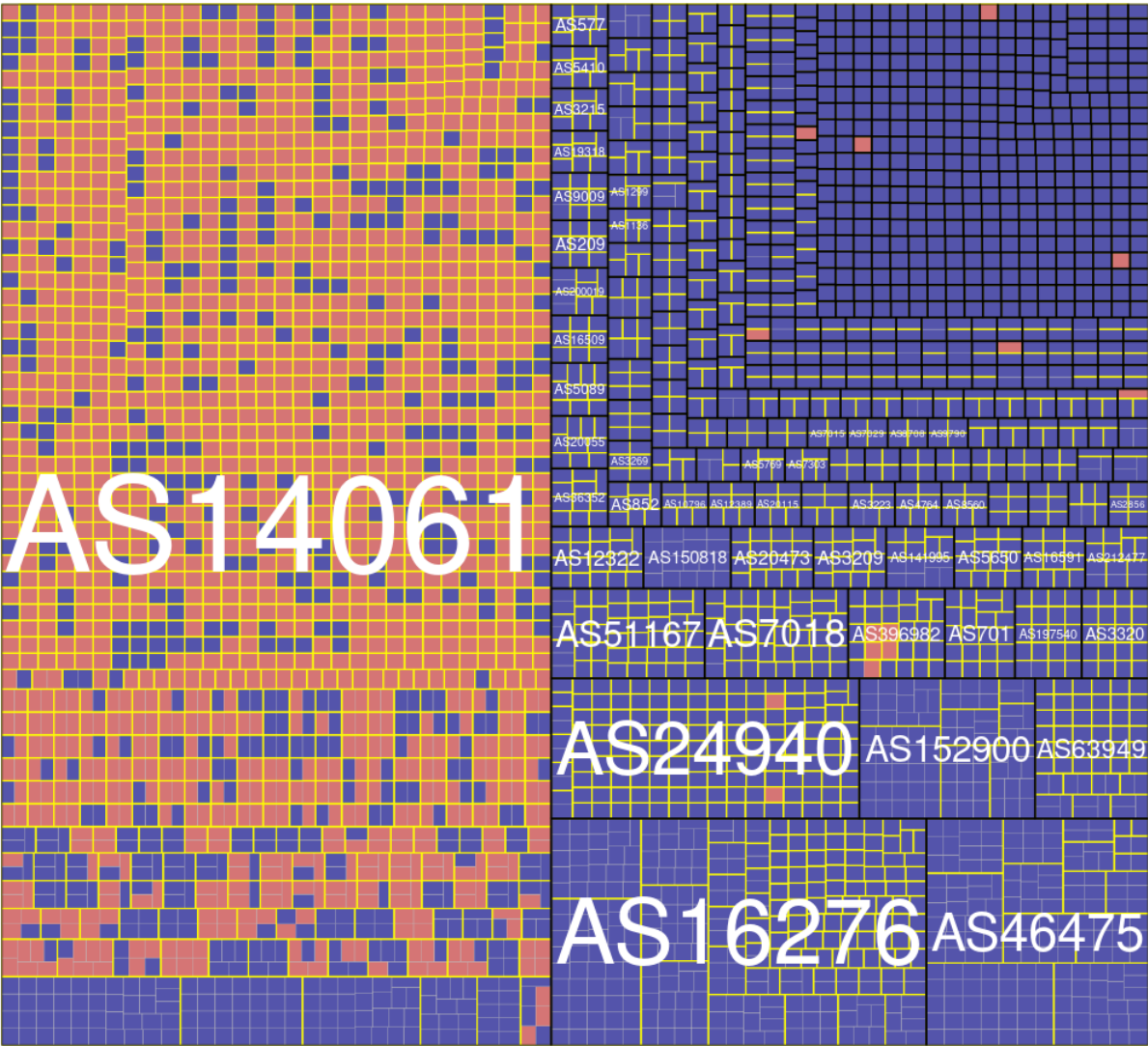
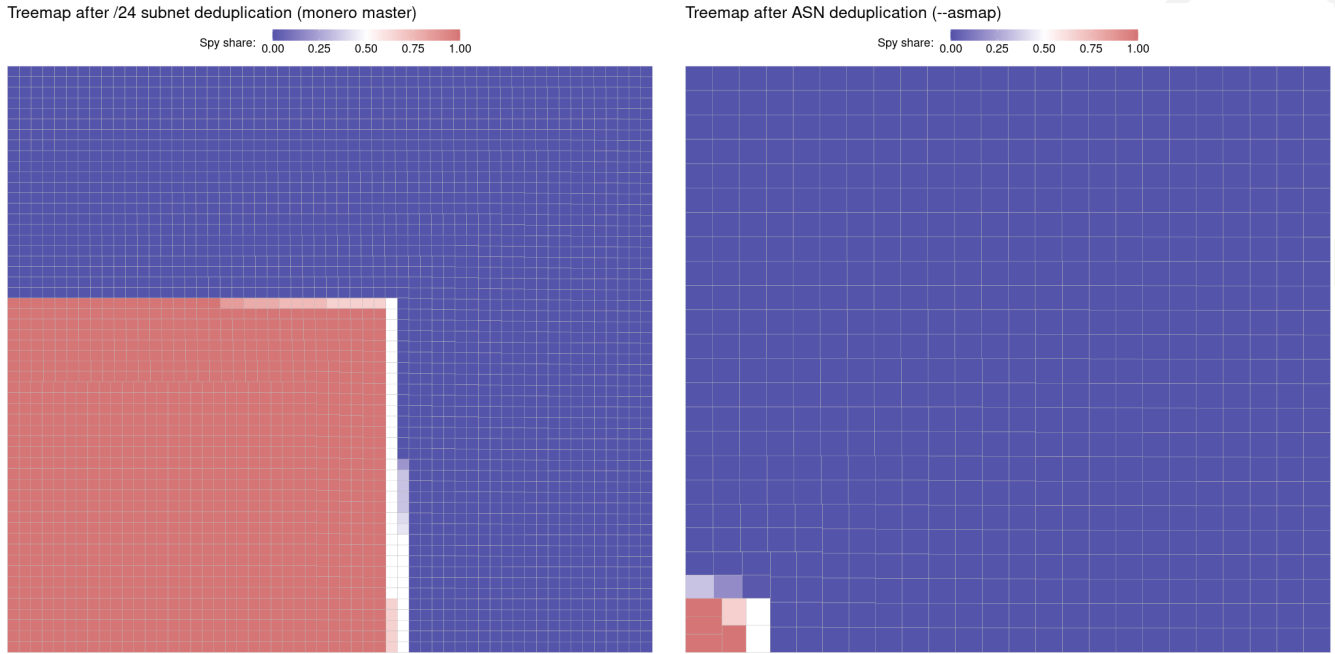


Figure 2: Groups after deduplication, October 2026: left, /24 deduplication (monero master); right, AS deduplication (-asmap)



97 4 Simulated effect on current suspected spy nodes

98 The simulation is the one in [Rucknium, 2025a], implemented by `gen.network()` in `xmrpeers` [Rucknium, 2025b],
 99 with one change: the subnet keys are replaced by a group key, so that the same function simulates /24 deduplication,
 100 AS deduplication, and the hybrid rule of Section 11. As in [Rucknium, 2025a], every honest node, reachable or not,
 101 fills 12 outbound slots and then replaces a random connection 12 times; suspected spy nodes make no outbound
 102 connections; unreachable nodes are added so that 80 or 90 percent of honest nodes are unreachable; and peer list
 103 size limits are ignored. The faster draw used here was checked against the literal shuffle-and-deduplicate draw of
 104 `gen.network()`, and both agree with the analytical selection probability within sampling error (Appendix A).

105 Table 2 gives the number of outbound connections, out of 12, from honest nodes to suspected spy nodes. With
 106 80 percent unreachable nodes, the mean in October 2026 is 4.35 (36.3 percent) with /24 deduplication and 0.13 (1.1
 107 percent) with AS deduplication. For the May 2025 data the means are 1.06 and 0.09; the /24 value reproduces the
 108 1.06 connections (8.8 percent) that [Rucknium, 2025a] reports for subnet deduplication on the same data, which
 109 validates the port of the simulation.

110 4.1 Inbound connections

111 Connections that no longer go to suspected spy nodes go to honest reachable nodes instead, and AS deduplication
 112 also moves connections from honest nodes in densely populated ASNs to honest nodes in sparsely populated ones
 113 (concern 4). Table 3 gives the distribution of inbound connections of honest reachable nodes, and Table 4 and Figure
 114 3 split it by the number of honest reachable nodes in the node’s AS. 62 percent of the ASes with honest reachable
 115 nodes contain only one. With AS deduplication every AS receives about the same share of inbound connections,
 116 which its nodes divide between them: a node receives about 801 inbound connections alone in its AS, 402 with
 117 one other honest node, 267 with two and 200 with three. With /24 deduplication most nodes receive about 151.
 118 The 764 nodes (28 percent) in ASes with up to 4 honest nodes therefore receive more inbound connections than
 119 today, and the others fewer. Figure 4 shows where the inbound connections go. Nodes alone in their AS are 13
 120 percent of honest reachable nodes; they receive 18 percent of all inbound connections with /24 deduplication and

Table 2: Summary statistics: Number of outbound connections, out of 12, to suspected spy nodes

Scenario	N honest nodes	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
Oct 2026, /24 dedup, 80% unreachable	18,622	0	3	4	4.35	5	11
Oct 2026, AS dedup, 80% unreachable	18,622	0	0	0	0.13	0	3
Oct 2026, hybrid, 80% unreachable	18,622	0	0	1	0.78	1	4
Oct 2026, /24 dedup, 90% unreachable	38,537	0	3	4	4.35	5	11
Oct 2026, AS dedup, 90% unreachable	38,537	0	0	0	0.13	0	3
Oct 2026, hybrid, 90% unreachable	38,537	0	0	1	0.78	1	4
May 2025, /24 dedup, 80% unreachable	21,192	0	0	1	1.06	2	7
May 2025, AS dedup, 80% unreachable	21,192	0	0	0	0.09	0	3
May 2025, hybrid, 80% unreachable	21,192	0	0	0	0.56	1	4
May 2025, /24 dedup, 90% unreachable	44,227	0	0	1	1.05	2	7
May 2025, AS dedup, 90% unreachable	44,227	0	0	0	0.09	0	3
May 2025, hybrid, 90% unreachable	44,227	0	0	0	0.56	1	4

Table 3: Summary statistics: Number of inbound connections to honest reachable nodes

Scenario	N honest reachable nodes	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
Oct 2026, /24 dedup, 80% unreachable	2,690	0	24	67	52.94	75.75	110
Oct 2026, AS dedup, 80% unreachable	2,690	0	1	7	82.19	105	452
Oct 2026, hybrid, 80% unreachable	2,690	0	9	81	77.64	140	190
Oct 2026, /24 dedup, 90% unreachable	2,690	0	49.25	142	109.58	155	198
Oct 2026, AS dedup, 90% unreachable	2,690	0	2	14	170.04	214	877
Oct 2026, hybrid, 90% unreachable	2,690	0	20	165	160.69	291	362
May 2025, /24 dedup, 80% unreachable	2,764	4	79	87	83.87	93	121
May 2025, AS dedup, 80% unreachable	2,764	0	7	37	91.32	133.25	372
May 2025, hybrid, 80% unreachable	2,764	4	75	96	87.71	106	140
May 2025, /24 dedup, 90% unreachable	2,764	11	171	182	175.16	191	227
May 2025, AS dedup, 90% unreachable	2,764	0	14	78	190.59	256.5	738
May 2025, hybrid, 90% unreachable	2,764	13	157	204	182.99	220	272

121 62 percent with AS deduplication. Nodes in ASes with ten or more honest nodes are 63 percent of the nodes and
 122 receive 51 percent and 4 percent. The live instances of Section 9 show the same shift on the real network: of their
 123 honest outbound peers, 11 percent were alone in their AS for default instances and 46 percent for instances with
 124 -asmap, and 53 and 13 percent were in ASes with ten or more honest nodes (92 and 132 peers; replication code:
 125 live_peer_classes.py). The live shift is smaller than simulated because real nodes choose from peer lists in which
 126 large-AS nodes are over-represented. The cause is the group draw: a node alone in its AS is drawn as often as a
 127 whole AS of hundreds of nodes. An adversary node alone in its AS is drawn as often too, which is the AS-distinct
 128 strategy analysed in Section 5.

Table 4: Inbound connections of honest reachable nodes by the number of honest reachable nodes in their AS
October 2026, 90 percent unreachable. Median: inbound connections per node.
Share: percent of all inbound connections of honest reachable nodes

Honest nodes in AS	ASes	Nodes (pct)	/24: median	AS: median	/24: share (pct)	AS: share (pct)
1	357	357 (13)	151	801	18.2	62.3
2	90	180 (7)	148	402.5	8.7	15.8
3	37	111 (4)	150	267	5.5	6.5
4	29	116 (4)	150	200	5.5	5.1
5-9	37	239 (9)	150	120	11.5	6.5
10-49	17	328 (12)	149	40.5	15.2	2.9
50-199	4	445 (17)	26	6	10.5	0.7
200+	2	914 (34)	71	1	24.9	0.2

Figure 3: Median inbound connections per honest reachable node, by the number of honest reachable nodes in its AS, October 2026, 90 percent unreachable nodes

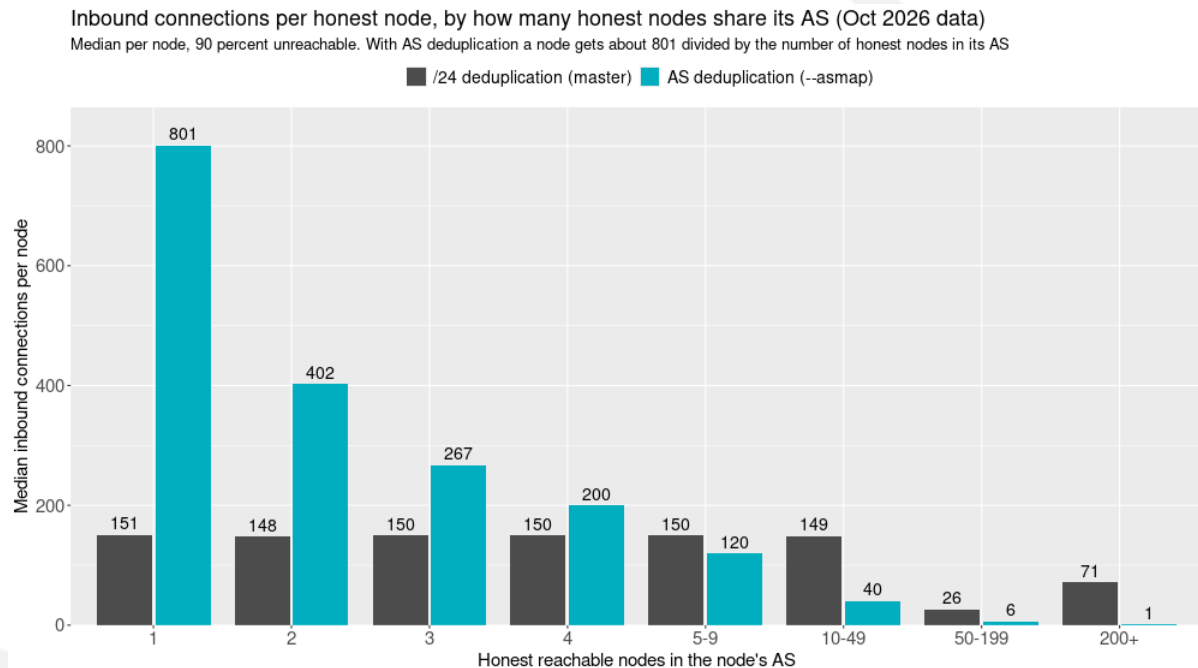
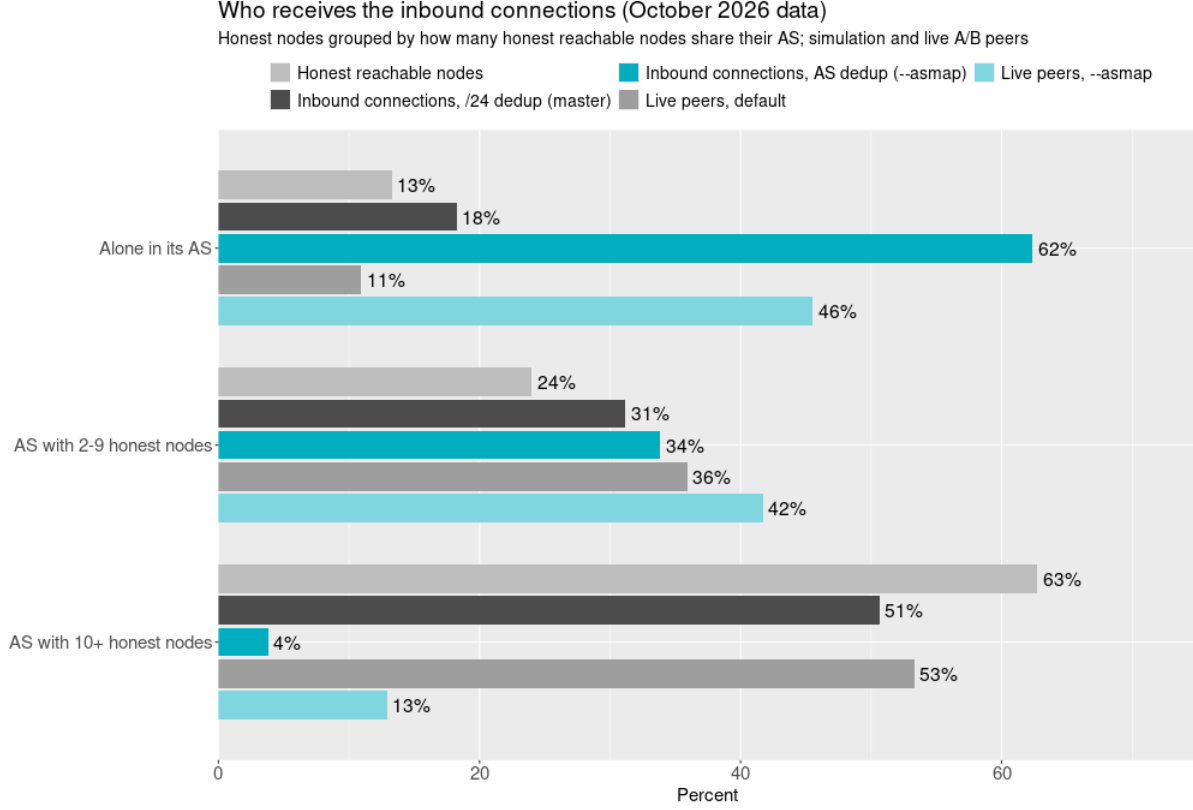


Figure 4: Share of honest reachable nodes and share of their inbound connections, by the number of honest reachable nodes in the node’s AS, October 2026, 90 percent unreachable nodes



5 Protocol-adversary interaction as a game

The simulation above holds the adversary’s behaviour fixed. As in Section 4 of [Rucknium, 2025a], we now let the adversary respond, and ask under what conditions AS deduplication is better for honest nodes than /24 deduplication. We make the assumptions of [Rucknium, 2025a]: the privacy impact is the probability that a single draw selects an adversary node; costs are linear in the number of nodes; and the adversary uses one strategy at a time. In addition we assume that the adversary does not place nodes in groups that contain honest nodes, which is the adversary’s best use of a node when every group costs the same and the adversary obtains less than about half of the draws (Appendix B).

The protocol chooses /24 deduplication (24) or AS deduplication (A). The adversary chooses between:

- *bulk*: nodes in distinct /24 subnets inside few ASes, for example virtual servers from one cloud provider, at price w_{24} per node and occupying m ASNs;
- *AS-distinct*: one node in each of many ASes, at price w_A per node. Each such node is also in a distinct /24 subnet.

Let H_{24} and H_A be the numbers of distinct /24 subnets and ASNs that contain at least one honest reachable node, and b the adversary’s budget. Following the derivation of equations (1) and (2) of [Rucknium, 2025a]:

$$p_{24,bulk} = \frac{b}{w_{24}H_{24} + b} \quad p_{24,AS} = \frac{b}{w_A H_{24} + b} \quad (1)$$

$$p_{A,bulk} \leq \frac{m}{H_A + m} \quad p_{A,AS} = \frac{b}{w_A H_A + b} \quad (2)$$

142 $p_{A,\text{bulk}}$ does not depend on the budget: however many nodes the adversary puts in its m ASes, they are m groups.
 143 **Proposition 1.** *Assume $w_{24} \leq w_A$, so that the bulk strategy is the adversary’s best response to /24 deduplication.*
 144 *The probability that a draw selects an adversary node is lower under AS deduplication against the AS-distinct strategy*
 145 *than under /24 deduplication against the bulk strategy, $p_{A,\text{AS}} < p_{24,\text{bulk}}$, if and only if*

$$\frac{w_A}{w_{24}} > \frac{H_{24}}{H_A}. \quad (3)$$

146 The proof is the same as for inequality (3) of [Rucknium, 2025a]. For a fixed price, AS deduplication is worse
 147 against the AS-distinct strategy than /24 deduplication is, $p_{A,\text{AS}} > p_{24,\text{AS}}$, because $H_A < H_{24}$. This is concern 2:
 148 honest nodes concentrated in a few server-provider ASes count as few groups, and each adversary node in its own
 149 AS counts as a full group. Inequality (3) quantifies it. With the October 2026 data $H_{24}/H_A = 1,983/573 = 3.46$,
 150 and with the May 2025 data $2,647/794 = 3.33$. If the adversary can find as many distinct ASes as it has nodes, AS
 151 deduplication is the better default only if a node in a distinct AS costs about three and a half times as much as a
 152 node in a distinct /24 subnet. The price survey and Section 5.2 below show that the price condition fails, and that
 153 the number of available ASes is what limits the adversary.

154 5.1 Prices

155 **Bulk.** w_{24} is low for the bulk strategy observed in October 2026. The cheapest virtual server of the provider that
 156 hosts 98.9 percent of the suspected spy nodes, AS14061, costs USD 4.00 per month and includes a public IPv4
 157 address [DigitalOcean, 2026]. The 1,279 suspected spy nodes in that AS are in 1,132 distinct /24 subnets in total
 158 (Table 1), so distinct /24 subnets come at almost no premium.

159 **Own ASN.** An adversary can register its own ASN, which costs EUR 50 per year at the RIPE NCC plus
 160 EUR 1,800 per year for an LIR account [RIPE NCC, 2025], and it must announce at least a /24 subnet, for which
 161 [Rucknium, 2025a] cites leasing prices of USD 118 to 250 per month. This is about thirty times the price of a USD
 162 4 virtual server, well above the threshold (3).

163 **Rented servers in other ASes.** The cheaper way to occupy a distinct AS is to rent a server from another
 164 hosting provider. Table 5 lists the cheapest plan with a public IPv4 address of 9 hosting providers that host
 165 honest reachable Monero nodes in the October 2026 data, taken from their pricing pages [Hosting providers, 2026,
 166 DigitalOcean, 2026], with EUR prices converted at the European Central Bank reference rate [European Central Bank, 2026].
 167 They cover 9 distinct ASNs, because one provider (Contabo) operates three regional ASNs. All 9 cost between
 168 USD 3.00 and 6.72 per month, at most 1.68 times the USD 4.00 bulk price, far below the threshold of 3.46. Price
 169 alone therefore does not protect AS deduplication: an adversary can obtain distinct ASes at nearly the bulk price.
 170 Pricing pages of some other providers that host honest nodes (Vultr, Akamai/Linode) refused automated access,
 171 and Hetzner’s page renders its prices with JavaScript, so they are not in the table; Hetzner charges EUR 0.50 per
 172 month extra for a primary IPv4 address [Hetzner, 2026].

173 What limits the adversary is the *number* of ASes in which it can rent servers. A spy node needs a public IPv4
 174 address that accepts inbound connections, which in practice means a hosting provider rather than a residential
 175 network. The X4BNet community list of datacenter ASNs [X4BNet, 2026] contains 933 ASNs. Of the 573 ASNs
 176 with honest reachable nodes in October 2026, 99 are on that list and hold 1,633 of the 2,690 honest nodes; the other
 177 474 are mostly residential and access networks.

178 5.2 Simulated adversary response

179 The payoffs above use a single draw. To include the 12 outbound slots and the churn, the simulation of the
 180 previous section was repeated with the observed suspected spy nodes replaced by synthetic adversary fleets. Only
 181 the honest reachable nodes are simulated as originators; unreachable honest nodes would add originators with the
 182 same distribution of outbound connections.

Table 5: Cheapest plan with a public IPv4 address, by hosting provider

Prices from the providers’ pricing pages, accessed 2026-10-05. EUR converted at the ECB reference rate of 2026-10-05 (1.1204 USD/EUR)

Provider	ASN	Plan	List price	USD/month	Honest nodes in ASN
InterServer	AS19318	1 Slice	3.00 USD	3.00	8
DigitalOcean	AS14061	Basic Droplet 512 MiB 1 vCPU 10 GiB	4.00 USD	4.00	630
OVHcloud	AS16276	VPS-1	4.54 USD	4.54	284
Contabo	AS51167	Cloud VPS	4.40 EUR	4.93	46
Contabo Asia	AS141995	Cloud VPS	4.40 EUR	4.93	14
Contabo US	AS40021	Cloud VPS	4.40 EUR	4.93	5
Amazon Lightsail	AS16509	Linux bundle with public IPv4	5.00 USD	5.00	9
IONOS	AS8560	VPS S+	6.00 USD	6.00	7
AlexHost	AS200019	U1	6.00 EUR	6.72	9

Table 6: Mean outbound connections, out of 12, to adversary nodes when the adversary responds

Data	Adversary	Adversary nodes	/24 dedup	AS dedup	Hybrid
Oct 2026	Observed suspected spy nodes	1293	4.38	0.13	0.80
Oct 2026	AS-distinct, $w_A/w_{24} = 1$	1293	4.71	8.31	5.32
Oct 2026	AS-distinct, $w_A/w_{24} = 2$	646	2.95	6.34	3.48
Oct 2026	AS-distinct, $w_A/w_{24} = 3.46$	373	1.90	4.73	2.34
Oct 2026	AS-distinct, $w_A/w_{24} = 5$	258	1.37	3.70	1.75
Oct 2026	AS-distinct, $w_A/w_{24} = 10$	129	0.71	2.22	0.93
May 2025	Observed suspected spy nodes	1843	1.05	0.09	0.56
May 2025	AS-distinct, $w_A/w_{24} = 1$	1843	4.92	8.37	5.15
May 2025	AS-distinct, $w_A/w_{24} = 2$	921	3.07	6.41	3.32
May 2025	AS-distinct, $w_A/w_{24} = 3.46$	532	1.99	4.81	2.20
May 2025	AS-distinct, $w_A/w_{24} = 5$	368	1.47	3.80	1.63
May 2025	AS-distinct, $w_A/w_{24} = 10$	184	0.81	2.31	0.88

Price premium. The fleet is replaced by $\lfloor n_{\text{spy}} w_{24}/w_A \rfloor$ nodes, each in its own new AS and /24 subnet, for price premiums w_A/w_{24} of 1, 2, 3.46, 5 and 10, where n_{spy} is the number of suspected spy nodes in the dataset (Table 6; replication code: `response_sim.R`). With no premium, the adversary obtains 8.31 of 12 connections under AS deduplication, compared with 4.71 under /24 deduplication (October 2026 data; May 2025: 8.37 and 4.92). This confirms concern 1 under the assumption that the adversary can find as many distinct ASes as it has nodes.

Limited supply of ASes. The more realistic constraint is the number of rentable ASes, K . The fleet of 1,293 nodes, each in its own /24 subnet, is spread evenly over K datacenter ASNs at an equal price per node, which Table 5 shows is approximately true for the cheapest providers. K is set to the 9 ASNs of Table 5, to the 99 datacenter ASNs that host honest reachable nodes, to all 933 listed datacenter ASNs, and to intermediate values made of the 99 plus randomly chosen other listed ASNs (replication code: `supply_sim.R`). Adversary nodes in an AS that also hosts honest nodes share that AS’s group, as in the real network. Table 7 and Figure 5 give the results. AS deduplication gives fewer connections to the adversary than /24 deduplication as long as the adversary can rent servers in fewer than about 370 ASes: 0.15 connections with $K = 9$, 1.61 with $K = 99$, against 4.71 with /24 deduplication. With $K = 933$ the adversary obtains 7.49 connections, more than with /24 deduplication. The break-even agrees with $k^* = 374$ from the single-draw model of Section 5. Repeated with 6 further random seeds

Table 7: Mean outbound connections, out of 12, to adversary nodes spread over K ASNs

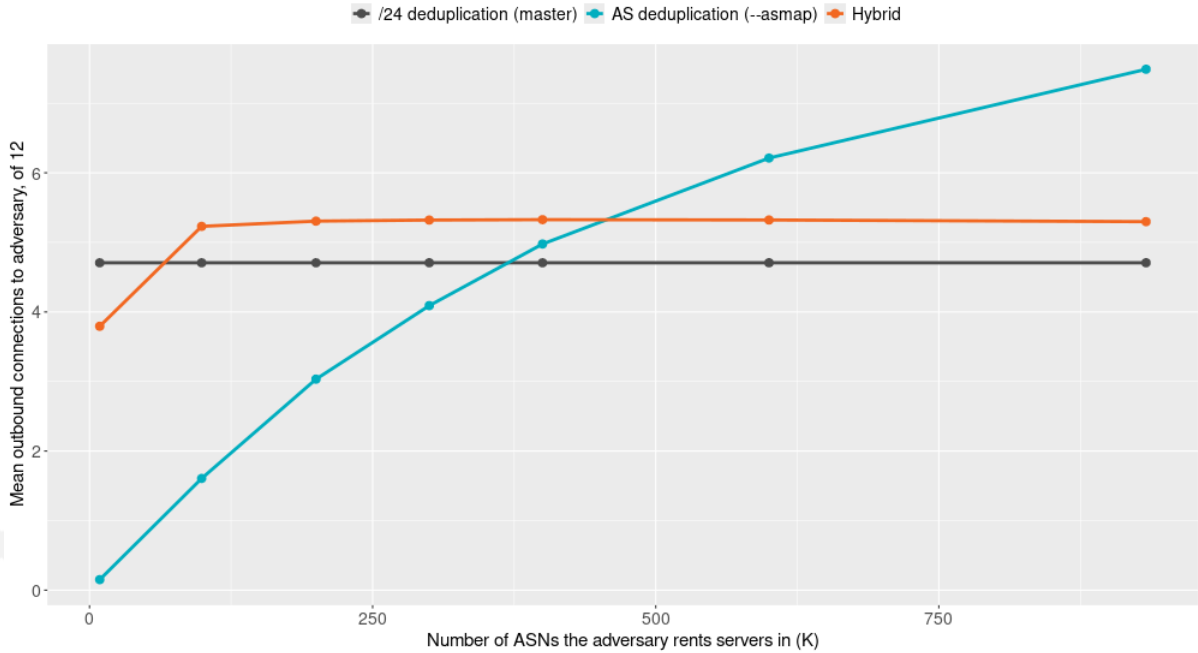
K	/24 deduplication (master)	AS deduplication (-asmap)	Hybrid
9	4.71	0.15	3.79
99	4.71	1.61	5.23
200	4.71	3.03	5.31
300	4.71	4.09	5.32
400	4.71	4.98	5.33
600	4.71	6.21	5.32
933	4.71	7.49	5.30

on the four vantage machines, the break-even is 377 ASes (standard deviation 5, range 369 to 382). If a node in another AS costs more than a bulk node, the adversary can afford fewer nodes, but under AS deduplication only the number of ASes it occupies matters: with a price ratio of 1.5 and 1.75 the break-even, measured against the full bulk fleet under /24 deduplication, is 383 and 398 ASes (replication code: `breakeven.R`).

Figure 5: Adversary connections when the adversary spreads over K rentable ASNs

Adversary spreads the observed fleet over K rentable ASNs (October 2026 data)

1,293 adversary nodes, each in its own /24, split evenly over K datacenter ASNs



5.3 Opt-in

If few nodes use `-asmap`, the adversary's best response is to the protocol that most nodes use, /24 deduplication, so it keeps the bulk strategy. Nodes that opt in then face the observed fleet, against which AS deduplication gives 0.13 connections to suspected spy nodes out of 12, against 4.35 with /24 deduplication (Table 2). The other nodes are unaffected, apart from the change in inbound connections caused by the opted-in nodes. The result therefore supports AS deduplication as an opt-in option.

The argument assumes that few nodes opt in. The MRL ban list is also opt-in, yet 72 percent of honest reachable nodes block it: their sampled white lists contain no ban-listed address, against a median of 14 percent for the other

nodes (replication code: `ban_adoption.py`; the method cannot distinguish the MRL list from a DNS blocklist that covers the same addresses). Because each node selects its outbound peers by its own rule, the number of ASes the adversary needs to offset AS deduplication, about 370, does not depend on how many nodes opt in, and nodes that do not opt in are unaffected. Adoption changes the adversary’s gain from spreading. If as many nodes opted in as block the ban list, spreading would win back most of the connections that AS deduplication removes, and for those nodes opt-in would become the default case of Section 6. The crawler measures the number of ASes the adversary uses continuously [plowsof, 2026a]. A rise from the current 10 ASes towards the hundreds would be the signal to advise nodes to turn the option off.

6 Can an adversary rent servers in about 370 ASes?

Section 5.2 reduces the question of a default to one number: the number K of distinct ASes in which an adversary can rent servers with inbound-reachable public IPv4 addresses. This section collects the available evidence and estimates K with a sample survey.

6.1 What adversaries have done

Every suspected spy fleet observed so far uses few ASes. The MRL ban list [Rucknium, 2024], version 2, covers 4,001 IPv4 addresses in 9 ASNs. The fingerprint-flagged reachable nodes of October 2026 are 1,293 addresses in 10 ASNs, 98.9 percent of them in one. A peer-list poisoning pool observed by the crawler is hosted by one proxy provider (PEG Tech); all addresses gossiped from that provider’s ASNs, 1,925 in total, fall in 3 ASNs. The hosting provider is not necessarily the operator. These adversaries chose bulk strategies against the status quo and /24 deduplication, which is their best response to those rules (Section 5). Their behaviour shows what is cheapest today, not what they could do if AS deduplication became the default.

6.2 Where outsiders run public servers

A lower bound on K is the number of ASes in which people other than the network operator already run inbound-reachable servers. Three networks give such counts, with addresses mapped to ASNs by the same asmap:

- Tor relays with an IPv4 address [The Tor Project, 2026]: 977 ASes, of which 173 are on the X4BNet datacenter list [X4BNet, 2026].
- Reachable Bitcoin nodes with an IPv4 address [Bitnodes, 2026]: 1,230 ASes, of which 132 are on the list.
- Reachable Monero nodes in October 2026: 573 ASes, of which 99 are on the list.

Together they cover 1,978 ASes, 233 of them on the datacenter list. The Monero crawler also records every address that peers gossip: since 2026-09-28, the four vantage points saw 98,483 distinct public IPv4 addresses in 4,953 ASNs, of which 4,018 addresses in 604 ASNs were reachable [plowsof, 2026a].

The 233 datacenter-listed ASes are fewer than 370, but the remaining 1,745 ASes are not all residential. The datacenter list is community-maintained and incomplete: for example, AS40021 (Contabo’s US network, Table 5) and AS210083 (Privex), both of which host honest reachable Monero nodes, are not on it. Counting therefore does not decide whether K exceeds 370.

6.3 Survey of rentable ASes

K was estimated by a random-sample survey of the 1,978 ASes that host public servers (replication code: `survey_sample.py`, `survey_fetch.py`, `survey_verdicts.py`, `survey_estimate.py`; saved copies of all pages in `data/survey/pages`).

1. A simple random sample of $n = 100$ ASes was drawn with a fixed seed.

Table 8: Sampled ASes classified as rentable

IPv4: incl. = included, add-on = priced add-on included in the price, n/s = not stated. Src: P = provider page, S = third-party listing, T = bgp.tools tag only

ASN	Network	Evidence	USD/mo	IPv4	Src
AS216071	VDSINA	VDS/VPS from USD 2.10/month	2.1	n/s	P
AS215467	Skhron	VPS EUR 1.69/month, IPv4 add-on EUR 0.65/m	2.62	add-on	P
AS7489	HostUS	KVM VPS, 512 MB, USD 35/year, 1 IPv4	2.92	incl.	S
AS202053	UpCloud	Starter cloud server from USD 3.5/month, I	3.5	incl.	S
AS51290	HosTeam	VPS Bronze 15.00 PLN/30d net, IPv4 + IPv6	3.84	incl.	P
AS58212	Dataforest	self-service cloud 'Seeds' from EUR 4.07/m	4.56	n/s	S
AS215590	DpkgSoft International Lim	virtual servers from 459 RUB/month	5.4	n/s	P
AS34700	Maxnet Ukraine	VPS 250 UAH/month, 1 IPv4 address	5.55	incl.	P
AS40244	Turnkey Internet	VPS pro.sm.amd USD 10/month	10.0	n/s	P
AS206264	Amarutu Technology Ltd.	VPS Medium Risk A USD 39.99/month	39.99	n/s	P
AS7296	Alchemy Communications, In	bare metal servers, private cloud, colocat			S
AS21581	M5 Computer Security / M5H	dedicated servers, bare metal, cloud insta			P
AS56030	Voyager Internet	cloud servers / virtual data centre, coloc			P
AS42675	Obehosting	dedicated servers and colocation, Stockhol			S
AS16302	SuomiCommunications	bgp.tools tag Server Hosting			T
AS12310	iNES Group	VPS and dedicated servers, price on reques			S
AS35366	ISPpro Internet KG	vServer and dedicated root servers			S
AS208208	CYBERSE	business cloud from 79 EUR/month, colocati			P
AS200303	LUMASERV	vServer / cloud computing			P
AS42532	Veesp SIA	VPS and dedicated servers			P
AS200950	Calibour Limited	VPS and dedicated servers			P
AS45899	VNPT	bgp.tools tag Server Hosting			T
AS399122	VPS House Technology Group	VPS, cloud and dedicated servers			P
AS49791	3HCLOUD	shared vCPU server USD 1.5/month, IPv4 pri		extra	P

2. Each AS was matched to its PeeringDB record (name, website and network type) [PeeringDB, 2026] and to the bgp.tools tags “Server Hosting”, “DSL” and “Mobile” [bgp.tools, 2026]. bgp.tools defines Server Hosting as a network that “provides commercial hosting, colocation, VPS, etc services (other people can buy machines and exit traffic from this network)”, which is the property that matters here.

3. The website of each AS was fetched and searched, in several languages, for offers of virtual or dedicated servers and their monthly prices. Where a site blocked automated access, a third-party listing was used and marked as secondary evidence.

4. Each AS was classified as *rentable* if it carries the Server Hosting tag or a server offer was found, *not rentable* if it carries only access-network tags or is registered as a residential, education, research or media network and no server offer was found, and *unknown* otherwise. Found evidence overrides tags.

Of the 100 sampled ASes, 24 are rentable, 47 are not rentable and 29 are unknown (Appendix D). Table 8 lists the rentable ones. A price was found for 10 of them, and 8 of those cost USD 7 per month or less, the range of Table 5; for 5 of the 8 the page states that an IPv4 address is included or gives its price. Prices in other currencies were converted at official reference rates [European Central Bank and Bank of Russia and National Bank of Ukraine, 2026, European Central Bank, 2026].

With 95 percent Wilson confidence intervals, scaled to the 1,978 ASes of the frame:

- rentable ASes: 475 (330 to 657); 1,048 (856 to 1,236) if every unknown AS is rentable;

- ASes confirmed rentable at USD 7 per month or less: 158 (81 to 297), of which 99 (43 to 221) with an IPv4 address stated in the price.

The second estimate is a lower bound: a price was found for only 10 of the 24 rentable ASes in the sample, and 8 of those 10 are cheap. The global estimate assumes that every AS with the Server Hosting tag is rentable. Because tagged ASes were classified as rentable by rule, the survey does not test this assumption; for 10 of the tagged or otherwise rentable ASes a priced offer was found.

The frame contains only ASes in which Tor, Bitcoin or Monero servers already run, and an adversary is not limited to them. `bgp.tools` tags 2,257 ASes as Server Hosting. In the sample the tag identified 20 of the 24 rentable ASes, a recall of 0.83, which suggests that about 2,719 ASes offer server hosting worldwide.

Result. The number of ASes in which servers can be rented is larger than the break-even of about 370: (330 to 657) within the frame alone and about 2,719 worldwide. The number confirmed to be as cheap as the bulk strategy is lower, 158 within the frame, but that is a lower bound and 8 of the 10 rentable ASes with a price were that cheap. Price and supply therefore do not appear to prevent an adversary from spreading over more than 370 ASes. The constraints that remain are operational: an adversary that uses 370 providers manages 370 accounts, payment methods and abuse desks, and each provider can suspend its servers. These costs are not measured here.

7 A complementary defence: AS-diverse Dandelion++ stems

The privacy loss that spy nodes cause is concentrated in the stem phase of Dandelion++, in which a node forwards its own transactions to one of its stem peers [Fanti et al., 2018]. Monero chooses `CRYPTONOTE_DANDELIONPP_STEMS = 2` stems per epoch uniformly at random, without replacement, from the node’s outbound connections (`connection_map` in `src/net/dandelionpp.cpp`). We consider a change to the stem choice only: draw the stems uniformly over the ASes of the outbound connections, then a connection uniformly within the AS. Outbound peer selection, and therefore the network graph and inbound load, are unchanged.

Against a bulk fleet, the connections that a node has to the fleet’s few ASes collapse into few stem candidates. Against an adversary with one node per AS, every adversary connection remains its own candidate, and only honest connections that share an AS are merged, so the adversary gains little. Table 9 gives the probability that a stem is an adversary node, computed exactly for each honest reachable node from its simulated outbound connections (replication code: `stem_sim.R`). With /24 deduplication and the observed suspected spy nodes it falls from 36.5 to 12.3 percent per stem, and the probability that either of the two stems is an adversary node falls from 59.7 to 24.3 percent. For a node that does not use the ban list it falls from 37.5 to 13.9 percent, and for the May 2025 data from 8.8 to 6.2 percent. Against one adversary node per AS (1,293 ASes, none shared with honest nodes), the worst case for every rule considered here, it rises from 39.2 to 42.5 percent.

Figure 6 shows how quickly each rule loses its advantage as the adversary spreads. AS-diverse stems lose theirs early: spread over the 99 datacenter ASes that host honest nodes, the adversary already obtains 41.9 percent per stem, and further spreading adds less than one percentage point. AS deduplication of outbound connections keeps its advantage longer (13.4 percent at 99 ASes) but loses it at about 370 ASes (39.7 percent) and keeps deteriorating, to 69.3 percent. No rule considered here is better than the current one against every adversary placement: a rule that trusts AS diversity tells the adversary which placement it trusts. The rules differ in what they lose in their worst case. AS-diverse stems lose at most 3.3 percentage points against the current rule and gain 24.2 against the observed fleet; AS deduplication of outbound connections has no comparable bound.

Table 9: Dandelion++ stem exposure: percent probability that a stem is an adversary node

Stem: current = 2 stems uniformly from the outbound connections (src/net/dandelionpp.cpp). AS-diverse = stems drawn uniformly over the ASes of the outbound connections

Data	Adversary	Outbound	Spy/12	Stem cur.	Stem AS-div.
Oct 2026	Observed	AS	0.1	1.1	1.1
Oct 2026	Observed	/24	4.4	36.5	12.3
Oct 2026	Observed + ban-list nodes	AS	0.2	1.3	1.3
Oct 2026	Observed + ban-list nodes	/24	4.5	37.5	13.9
Oct 2026	Spread, 99 ASes	AS	1.6	13.4	13.4
Oct 2026	Spread, 99 ASes	/24	4.7	39.2	41.9
Oct 2026	Spread, 370 ASes	AS	4.8	39.7	39.7
Oct 2026	Spread, 370 ASes	/24	4.7	39.2	42.3
Oct 2026	Spread, 933 ASes	AS	7.5	62.4	62.4
Oct 2026	Spread, 933 ASes	/24	4.7	39.2	42.4
Oct 2026	One node per AS	AS	8.3	69.3	69.3
Oct 2026	One node per AS	/24	4.7	39.2	42.5
Oct 2026	10 largest honest ASes	AS	0.1	1.1	1.1
Oct 2026	10 largest honest ASes	/24	4.7	39.2	36.7
Oct 2026	30 largest honest ASes	AS	0.4	3.2	3.2
Oct 2026	30 largest honest ASes	/24	4.7	39.2	40.4
Oct 2026	Honest AS distribution	AS	2.7	22.8	22.8
Oct 2026	Honest AS distribution	/24	4.7	39.2	38.0
May 2025	Observed	AS	0.1	0.8	0.8
May 2025	Observed	/24	1.1	8.8	6.2

Figure 6: Probability that a Dandelion++ stem is an adversary node as the adversary spreads its nodes over more ASes, October 2026

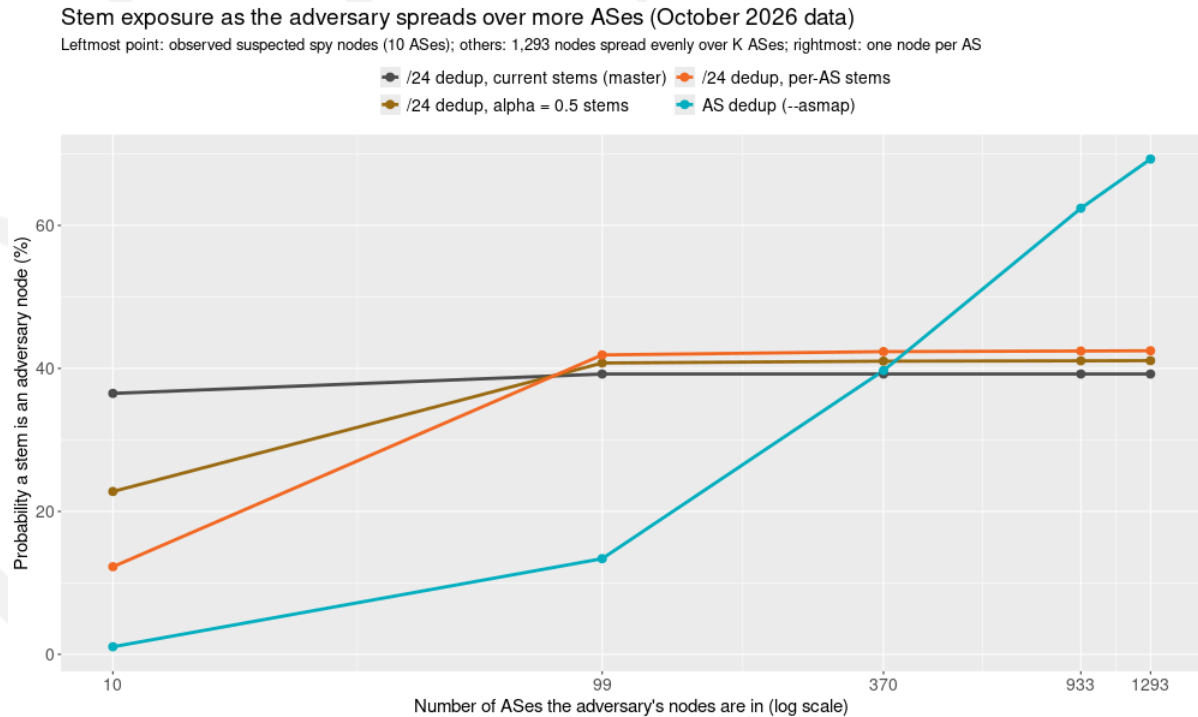


Table 10: Stem exposure (percent) with connections weighted by $n^{-\alpha}$, /24 deduplication, October 2026

Weight exponent α	One node per AS	Observed suspected spy nodes
0	39.2	36.5
0.25	40.2	29.5
0.5	41.1	22.8
0.75	41.8	17.0
1	42.5	12.3

Table 11: Expected stem traffic received per honest reachable node, /24 deduplication, October 2026

Sum over originators of the probability that the node is chosen as a stem. cur. = current rule, div. = AS-diverse. Alone: only honest reachable node in its AS

Adversary	Med. cur.	Med. div.	Max cur.	Max div.	Gini cur.	Gini div.	Alone cur.	Alone div.
One node per AS	0.75	0.64	1.92	2.03	0.29	0.33	0.83	0.89
Observed + ban-list	0.75	1.00	2.00	3.49	0.29	0.40	0.87	1.47
Observed	0.75	1.03	1.83	3.10	0.29	0.40	0.86	1.45

The trade-off can be tuned by weighting each connection by $n^{-\alpha}$, where n is the number of the node’s outbound connections in the same AS: $\alpha = 0$ is the current rule and $\alpha = 1$ the AS-diverse rule (Table 10). The worst-case cost grows slowly with α and the benefit against the observed fleet quickly: with $\alpha = 0.5$ the stem exposure is 22.8 percent against the observed fleet and 41.1 percent in the worst case, at most 1.9 percentage points above the current rule.

Stem load. AS-diverse stems move stem traffic towards honest nodes that are alone in their AS, the stem counterpart of concern 4. Table 11 gives the expected stem traffic each honest reachable node receives. Against the observed suspected spy nodes the median rises from 0.75 to 1.03, partly because traffic that went to spy nodes now goes to honest ones; the maximum rises from 1.83 to 3.10, the Gini coefficient from 0.29 to 0.40, and nodes alone in their AS receive 1.45 instead of 0.86. The concentration is moderate compared with the inbound concentration of AS deduplication of outbound connections (Section 4.1).

Adversaries aimed at the stem rule. An adversary that knows the stem rule might place its nodes in the ASes that hold the most honest nodes, so that honest connections are merged with its own. With the observed fleet size, concentrating in the 10 or 30 largest honest ASes, or copying the honest AS distribution, gives 36.7, 40.4 and 38.0 percent per stem (Table 9), all below the one-node-per-AS case of 42.5 percent.

AS-diverse stems are not part of pull request #11474. They need the asmap but no change to peer selection, so they could be used with /24 deduplication by every node that loads the map. Their effect on the anonymity guarantees of Dandelion++, which assume random stem choice among outbound connections, should be reviewed before they are adopted.

8 Selection from real peer lists

The simulations above follow [Rucknium, 2025a] in letting every node choose from all reachable nodes. Real nodes choose from a white list of at most 1,000 and a gray list of at most 5,000 addresses, built from the peer lists of their neighbours, and prefer recently seen white-list entries. That model predicted 30.8 percent of outbound connections to suspected spy nodes for the status quo in May 2025 [Rucknium, 2025a], while [Kopyciok et al., 2026] measured

328 6.69 to 7.78 percent on default nodes in June 2025. Absolute shares from the uniform model are therefore probably
329 too high.

330 To use real candidate pools, each of the 2,527 honest reachable nodes that completed a handshake was asked four
331 times, from the four vantage points, for its peer list; each response is a random sample of up to 250 white-list entries
332 (replication code: `pl_crawl.py`). On average 17.5 percent of a node’s entries are suspected spy nodes or reachable
333 ban-listed nodes, close to the 16.22 percent of peer-list entries that [Kopyciok et al., 2026] attribute to one AS; 28.5
334 percent are addresses not known to be reachable, which are treated as failed connection attempts. Each node then se-
335 lects outbound connections from its own list with the algorithm of `make_new_connection_from_peerlist()`: dedu-
336 plication by group, sorting by `last_seen`, the first 20 untried candidates, and the cubic preference for recent entries
337 of `get_random_index_with_fixed_probability()` (replication code: `pl_sim.py`). Only the white list is modelled;
338 monerod draws about 70 percent of outbound connections from it (`P2P_DEFAULT_WHITELIST_CONNECTIONS_PERCENT`).

339 Table 12 gives the results. With /24 deduplication and no ban list, 2.73 of 12 outbound connections go to
340 suspected spy nodes, fewer than in the uniform model, and AS deduplication reduces this to 0.71. In the spread case
341 the same list entries are relabelled so that every adversary node is in its own AS, the most favourable placement for
342 the adversary; AS deduplication then gives 3.35 connections against 2.73 with /24 deduplication, a smaller penalty
343 than in the uniform model.

344 **AS-aware peer lists.** Peer-list poisoning is the mechanism that gives the adversary its share of candidates
345 [Kopyciok et al., 2026], who suggest pruning duplicated subnets or ASes from peer lists. Bitcoin Core applies the
346 asmap to the same problem: it uses the AS to place addresses in its address manager’s buckets, which limits the
347 share of the address table that one AS can occupy [naumenkogs, 2019]. Monero’s white and gray lists have no such
348 limit. Capping each AS at 3.1 percent of a node’s list, the fraction of Bitcoin’s tried table that one group can occupy
349 (8 of 256 buckets [Bitcoin Core, 2026a]), with /24 deduplication otherwise unchanged, reduces the connections to
350 the observed suspected spy nodes from 2.73 to 0.66, and a 1 percent cap to 0.26. Against the spread case the cap
351 gives 3.21 and 3.59 connections, against 2.73, because capping the large honest ASes raises the adversary’s share of
352 what remains. The cap is applied here to a static list; a cap applied when entries are inserted, as in Bitcoin, may
353 behave differently and should be simulated with list dynamics.

354 9 Live measurement on the Monero network

355 All results so far are simulations. To measure the effect directly, unmodified `monerod` instances built from pull
356 request #11474 were run with `-no-sync`, 12 outbound and no inbound connections, and no ban list, on the 4
357 vantage machines: 12 instances with `-asmap=embedded` and 12 without, three of each per machine (replication code:
358 `ab_start_any.sh`, `ab_poll.py`, `ab_analyze.py`). Their outbound connections were recorded every 5 minutes and
359 classified with the October 2026 suspected spy set and the reachable ban-listed nodes. This is the configuration of
360 the default nodes measured by [Kopyciok et al., 2026].

361 After 8.9 hours, the default instances had 33.8 percent of their outbound connections (4.05 of 12) to suspected
362 spy nodes, and the instances with `-asmap` had 2.7 percent (0.32 of 12). The default instances’ peers came from
363 7.0 distinct ASes on average, the asmap instances’ from 12.0. With AS-diverse stems, the default instances’ stem
364 exposure would have been 13.8 percent; this is computed from their recorded connections, not measured on nodes
365 that use the rule.

366 Outbound connections of a node that does not sync change little: each instance had between 12 and 18 distinct
367 outbound peers over the whole run. The snapshots of one instance are therefore not independent, and the unit of
368 comparison is the instance. Between instances the default share ranged from 17 to 58 percent and the asmap share
369 from 0 to 8 percent; every default instance had a higher share than every asmap instance (exact one-sided Wilcoxon
370 rank-sum test over instance means, $p = 3.7 \times 10^{-7}$). The default share is closer to the uniform model than to the
371 real-peer-list simulation and higher than the 7.78 percent that [Kopyciok et al., 2026] measured in June 2025; the
372 two measurements classify different sets of suspected spy nodes, sixteen months apart.

Table 12: Outbound selection from real peer lists

Each honest reachable node selects from the white-list entries it returned in four handshakes. Spread: the same entries, one AS per adversary node. Cap: per-AS cap on list entries

Adversary	Ban	Strategy	List spy %	Spy/12	Stem cur. %	Stem AS-div. %
Observed	off	AS	17.5	0.71	5.9	5.9
Observed	off	AS + cap 1.0%	1.5	0.24	2.0	2.0
Observed	off	AS + cap 3.1%	3.9	0.42	3.5	3.5
Observed	off	/24	17.5	2.73	22.7	10.7
Observed	off	/24 + cap 1.0%	1.5	0.26	2.2	2.1
Observed	off	/24 + cap 3.1%	3.9	0.66	5.5	4.7
Observed	on	AS	14.7	0.47	3.9	3.9
Observed	on	AS + cap 1.0%	0.9	0.13	1.1	1.1
Observed	on	AS + cap 3.1%	2.4	0.27	2.3	2.3
Observed	on	/24	14.7	2.42	20.2	7.8
Observed	on	/24 + cap 1.0%	0.9	0.16	1.3	1.3
Observed	on	/24 + cap 3.1%	2.4	0.41	3.4	2.9
Spread	off	AS	17.5	3.35	27.9	27.9
Spread	off	AS + cap 1.0%	24.0	3.81	31.8	31.8
Spread	off	AS + cap 3.1%	20.9	3.49	29.1	29.1
Spread	off	/24	17.5	2.73	22.7	25.0
Spread	off	/24 + cap 1.0%	24.0	3.59	29.9	30.2
Spread	off	/24 + cap 3.1%	20.9	3.21	26.8	27.5
Spread	on	AS	14.7	2.91	24.3	24.3
Spread	on	AS + cap 1.0%	20.5	3.31	27.6	27.6
Spread	on	AS + cap 3.1%	17.7	3.00	25.0	25.0
Spread	on	/24	14.7	2.42	20.2	22.3
Spread	on	/24 + cap 1.0%	20.5	3.20	26.7	26.9
Spread	on	/24 + cap 3.1%	17.7	2.85	23.7	24.4

10 Related work

Eclipse attacks. Bitcoin Core’s asmap and address-manager bucketing respond to eclipse attacks that fill a node’s address tables [Heilman et al., 2015, Tran et al., 2020, naumenkogs, 2019]. Shi et al. eclipsed Monero nodes by populating their gray and white lists from many addresses and then forcing them to drop benign connections [Shi et al., 2025]. Their countermeasures include ignoring peer lists from incoming connections and limiting the records one connection can insert; the AS cap of Section 8 limits what one AS can insert instead. The double-spend form of their connection reset was fixed in v0.18.3.2 (pull request #9218, as reported in [Shi et al., 2025]).

Measurement of the Monero network. Cao et al. measured Monero’s peer-to-peer network and its peer lists [Cao et al., 2020]. Kopyciok, Schmid and Victor detected anomalous peers from five vantage nodes in June 2025 and attributed a self-promoting Sybil cluster to one AS [Kopyciok et al., 2026].

Anonymity of Dandelion++. Sharma, Gosain and Diaz model Dandelion and Dandelion++ with Bayesian inference and find that an adversary controlling 20 percent of nodes intercepts a similar fraction of transactions in Dandelion++ as in Dandelion, with a median anonymity set equivalent to about 32 possible originators [Sharma et al., 2023]. The adversary’s share of stem relays, Section 7, is therefore the measure that matters for privacy.

Comparison with published measurements. Table 13 sets this note’s measurements beside the published ones. The fleet’s share of peer-list entries agrees with [Kopyciok et al., 2026]. The fleet now in AS14061 is of similar size to the one they observed in AS54098, which filled seven /24 subnets, but it is spread over 1,132 /24 subnets, which /24 deduplication does not limit.

Table 13: Measurements in this note and in published work
Denominators and classifications differ between studies; see the text

Quantity	This note	Published
Simulation, May 2025 data, /24 deduplication	1.06 of 12 outbound connections to ban-listed nodes	1.06 of 12 (8.8%) [Rucknium, 2025a]
Non-standard nodes	1,293 of 3,983 reachable IPv4 nodes, Oct 2026	1,877 of 13,050 connected IPs (14.38%), Jun 2025 [Kopyciok et al., 2026]
Largest entity	1,279 nodes in AS14061	1,582 peers in AS54098 [Kopyciok et al., 2026]; 566 of 1,843 ban-listed nodes in AS54098, May 2025 [Rucknium, 2025b]
Fleet share of peer-list entries	17.5% (real peer lists)	16.22% for AS54098 [Kopyciok et al., 2026]
Default node’s outbound connections to the fleet	33.8% live; 22.7% real-list simulation; 36.3% uniform model	6.69% (AS54098) and 7.78% (all anomalous), measured [Kopyciok et al., 2026]; 30.8% before and 8.8% after /24 deduplication, modelled [Rucknium, 2025a]
Addresses available to the fleet	1,293 IPv4 addresses in 1,132 /24 and 77 /16 subnets	eclipse attack evaluated with 1,020 addresses [Shi et al., 2025]

11 A hybrid rule

/24 deduplication fails against the bulk strategy and AS deduplication against the AS-distinct strategy. A hybrid rule keeps /24 deduplication and in addition refuses an outbound candidate whose ASN already has an outbound connection, the per-AS connection limit proposed in Monero issue #7090 [keffnet, 2020]. It is not part of pull request #11474. Against the observed fleet it performs between the two rules (0.78 connections out of 12, Table 2). Against an adversary spread over many ASes it performs *worse* than /24 deduplication: 5.23 connections with $K = 99$ and 5.30 with $K = 933$, against 4.71 (Table 7). After an honest node connects to a large honest AS, the hybrid rule excludes all /24 subnets of that AS from later draws, while the adversary’s /24 subnets elsewhere remain. The hybrid rule is therefore not a candidate for a default.

12 Conclusions

1. Against the suspected spy nodes observed in October 2026, AS deduplication reduces outbound connections to them from 4.35 to 0.13 out of 12 in simulation, and from 33.8 to 2.7 percent on live nodes. As an opt-in option it gives this benefit without giving the adversary a reason to change strategy while few nodes use it. At the adoption of the ban list, 72 percent of reachable nodes, the adversary would have that reason, so the adversary’s AS count should be monitored.
2. As a default, AS deduplication is better than /24 deduplication only if the adversary cannot rent servers in more than about 370 ASes. Price does not prevent this, and an estimated 475 (330 to 657) ASes within the survey frame, and about 2,719 worldwide, rent servers. The evidence does not support AS deduplication as the default.
3. AS deduplication concentrates inbound connections on honest nodes alone in their AS: they are 13 percent of honest reachable nodes and would receive 18 percent of inbound connections today but 62 percent with AS deduplication.

4. AS-diverse Dandelion++ stems, with /24 deduplication unchanged, reduce stem exposure from 36.5 to 12.3 percent (real peer lists: 22.7 to 10.7). Their advantage disappears once the adversary spreads over about 99 ASes, but their worst-case loss is 3.3 percentage points (1.9 with $\alpha = 0.5$). No rule considered is better than the current one against every placement; this one has the smallest worst-case loss among those that help against the observed fleet. It is the candidate for a default, subject to review of its effect on the Dandelion++ anonymity analysis.
5. An AS cap on peer-list entries reduces connections to the observed fleet the most (2.73 to 0.66 of 12), but like AS deduplication it loses against a spread adversary (3.21 against 2.73). It needs a simulation with peer-list dynamics.
6. The uniform candidate model of [Rucknium, 2025a] gives higher absolute shares than real peer lists; the ranking of the rules is the same in both.
7. A hybrid of /24 deduplication and a per-AS connection limit is worse than /24 deduplication against a spread adversary.
8. 83.0 percent of the gossiped MRL ban-list addresses are reachable, in 68 /24 subnets, so /24 deduplication already limits them.

13 Limitations and further research

- The survey of rentable ASes classifies 29 of 100 sampled ASes as unknown and finds prices for 10 of 24 rentable ones; the operational cost of using hundreds of providers is not measured.
- The suspected spy classification of the October 2026 data is withheld; the May 2025 data reproduce the network simulations and the price-premium response with public data.
- The real-peer-list simulation models only the white list, treats entries not known to be reachable as failed attempts, applies the AS cap to a static list, and models the spread adversary by relabelling the observed list entries.
- The single-draw model ignores drawing without replacement and the fixed-probability preference for recently seen white list peers; the simulation ignores peer list size limits and treats every reachable node as equally likely to be in a peer list, as in [Rucknium, 2025a].
- The supply-constrained simulation spreads adversary nodes evenly over the ASes; price ratios up to 1.75 were simulated. The stem analysis assumes the adversary does not adapt its strategy to the stem rule beyond the strategies simulated.
- The asmap is refreshed when Monero releases. New prefixes are attributed to a neighbouring AS in the filled map until the next refresh [bitcoin-core/asmap-data, 2026].
- The live measurement has 12 instances per arm on four machines in three hosting providers' networks (OVH, Hetzner and Oracle Cloud), without inbound connections, over less than one day; outbound peers changed little during the run.
- AS-diverse stems were evaluated by simulation and by computation from recorded connections, not on nodes that implement the rule. Stem exposure is the probability for the first stem hop; multi-hop stem paths and the resulting anonymity set [Sharma et al., 2023] are not modelled.
- IPv6 and anonymity networks are not analyzed.

452 Code and data

453 A replication package contains all code and maps every table, figure and number to its script. The October 2026
454 data, from the Monero P2P Observatory crawl [plowsof, 2026a] and the live runs, are included with IP addresses
455 replaced by keyed hashes and subnets by order-preserving labels; with them, the stem, peer-list and live analyses
456 reproduce this note’s results exactly. Raw crawl records and the fingerprint classifier are available to the Monero
457 Research Lab on request.

458 Acknowledgements

459 The simulation, game-theoretic model and May 2025 data are Rucknium’s [Rucknium, 2025a, Rucknium, 2025b].
460 /24 deduplication was implemented by rrunner7 [rrunner7, 2025], a per-AS connection limit for Monero was
461 proposed by keffnet [keffnet, 2020] and AS grouping was first implemented by Malinero [Malinero, 2021], and the
462 asmap data are produced by the bitcoin-core/asmap-data contributors [bitcoin-core/asmap-data, 2026].

463 References

- 464 [Apostolaki et al., 2017] Apostolaki, M., Zohar, A., and Vanbever, L. (2017). Hijacking Bitcoin: Routing attacks
465 on cryptocurrencies. In *2017 IEEE Symposium on Security and Privacy (SP)*, pages 375–392.
- 466 [b10c, 2024] b10c (2024). One year update on LinkingLion. Accessed 2026-10-06.
- 467 [bgp.tools, 2026] bgp.tools (2026). Networks with the tag “server hosting” (vps), “dsl” and “mobile”. “This network
468 provides commercial hosting, colocation, VPS, etc services (other people can buy machines and exit traffic from
469 this network).” 2,257 networks. Accessed 2026-10-05.
- 470 [Bitcoin Core, 2026a] Bitcoin Core (2026a). `src/addrman.cpp`: `ADDRMAN_TRIED_BUCKETS_PER_GROUP{8}` (line 26);
471 `src/addrman_impl.h`: `ADDRMAN_TRIED_BUCKET_COUNT_LOG2{8}` (line 26). commit 09e22fbbb0ca.
- 472 [Bitcoin Core, 2026b] Bitcoin Core (2026b). `src/init.cpp`, `-asmap` option help text (line 560) and `src/net.cpp`,
473 one outbound peer per network group (line 2724). commit 09e22fbbb0ca, 2026-10-05.
- 474 [bitcoin-core/asmap-data, 2026] bitcoin-core/asmap-data (2026). Repository holding recently created ASMaps en-
475 coded for use in Bitcoin Core. File 2026/1788801420_asmap.dat (filled map; an `_unfilled` variant is published
476 alongside). MIT licence.
- 477 [Bitnodes, 2026] Bitnodes (2026). Latest snapshot of reachable Bitcoin nodes. Snapshot timestamp 1791231890.
478 Accessed 2026-10-05.
- 479 [Cao et al., 2020] Cao, T., Yu, J., Decouchant, J., Luo, X., and Verissimo, P. (2020). Exploring the monero peer-
480 to-peer network. In Bonneau, J. and Heninger, N., editors, *Financial Cryptography and Data Security*, pages
481 578–594. Springer International Publishing.
- 482 [DigitalOcean, 2026] DigitalOcean (2026). Droplet pricing. Cheapest Basic Droplet USD 4.00/month, includes a
483 public IPv4 address. Accessed 2026-10-05.
- 484 [European Central Bank, 2026] European Central Bank (2026). Euro foreign exchange reference rate, USD, 2026-
485 10-05: 1.1204.
- 486 [European Central Bank and Bank of Russia and National Bank of Ukraine, 2026] European Central Bank and
487 Bank of Russia and National Bank of Ukraine (2026). Reference exchange rates: 4.3795 pln/eur (ecb, 2026-
488 10-05), 84.9309 rub/usd (bank of russia), 45.0564 uah/usd (nbu, 2026-10-06).
- 489 [Fanti et al., 2018] Fanti, G., Venkatakrishnan, S. B., Bakshi, S., Denby, B., Bhargava, S., Miller, A., and
490 Viswanath, P. (2018). Dandelion++: Lightweight cryptocurrency networking with formal anonymity guaran-
491 tees. *Proc. ACM Meas. Anal. Comput. Syst.*, 2(2).

[fjahr, 2023] fjahr (2023). build: Embedded ASMap [3/3]: Build binary dump header file. Bitcoin Core pull request #28792, merged 2026-02-19.

[Heilman et al., 2015] Heilman, E., Kendler, A., Zohar, A., and Goldberg, S. (2015). Eclipse attacks on Bitcoin’s peer-to-peer network. In *24th USENIX Security Symposium*.

[Hetzner, 2026] Hetzner (2026). Primary IPs: Overview. “We charge EUR 0.50 / USD 0.60 per month (excl. VAT) for our Primary IPv4s.” Accessed 2026-10-05.

[Hosting providers, 2026] Hosting providers (2026). Pricing pages of interserver, digitalocean, ovhcloud, contabo, amazon lightsail, ionos and alexhost. URLs and quoted IPv4 terms in `data/prices/supply.csv`; page copies saved on access, 2026-10-05.

[IPtoASN, 2026] IPtoASN (2026). IP address to ASN database, combined IPv4 and IPv6 (`ip2asn-combined.tsv.gz`). Downloaded 2026-10-05; used for AS names only.

[keffnet, 2020] keffnet (2020). Per-AS connection limit. Monero issue #7090, opened 2020-12-07.

[Kopyciok et al., 2026] Kopyciok, Y., Schmid, S., and Victor, F. (2026). Friend or foe? anomalous peers in Monero’s P2P network. In *8th Conference on Advances in Financial Technologies (AFT 2026)*, volume 395 of *Leibniz International Proceedings in Informatics (LIPIcs)*, pages 31:1–31:22. CC-BY 4.0.

[Malinero, 2021] Malinero (2021). p2p: group peers by AS (bitcoin’s asmap). Monero pull request #7935, opened 2021-09-11, closed unmerged.

[naumenkogs, 2019] naumenkogs (2019). p2p: supplying and using asmap to improve IP bucketing in addrman. Bitcoin Core pull request #16702, merged 2020-01-29.

[#no-wallet-left-behind, 2026] #no-wallet-left-behind (2026). IRC log, 2026-10-05, 16:31–16:52 UTC.

[PeeringDB, 2026] PeeringDB (2026). Network records (`api/net`): name, website and network type. Accessed 2026-10-05.

[plowsof, 2026a] plowsof (2026a). Monero P2P observatory. Four-vantage crawl of reachable Monero nodes; snapshot used here taken 2026-10-05.

[plowsof, 2026b] plowsof (2026b). p2p: group peers by AS (bitcoin’s asmap). Monero pull request #11474, opened 2026-10-03. Embedded-map variant: branch `test-asmap`, commit 82461b4b6.

[rbrunner7, 2025] rbrunner7 (2025). p2p: Improved peer selection with /24 subnet deduplication to disadvantage ‘spy nodes’. Monero pull request #9939, merged 2025-09-28.

[RIPE NCC, 2025] RIPE NCC (2025). RIPE NCC charging scheme 2026. Technical Report ripe-848, RIPE Network Coordination Centre. EUR 1,800 per LIR account, EUR 50 per ASN assignment. Accessed 2026-10-05.

[Rucknium, 2024] Rucknium (2024). MRL recommendation: Ban spy node IP addresses from connecting to your node. monero-project/meta issue #1124, opened 2024-12-11.

[Rucknium, 2025a] Rucknium (2025a). Subnet deduplication for monero node peer selection. Research bulletin draft v0.3, Monero Research Lab.

[Rucknium, 2025b] Rucknium (2025b). xmrpeers: Analyze monero peer-to-peer network data (r package). Datasets `good_peers` and `ban_list_v1`; function `gen.network()` in `R/peer-selection.R`.

[selsta, 2026] selsta (2026). `js/RemoteNodes.js` in monero-gui pull request #4601 (replace simple mode bootstrap with remote nodes). commit 7db60a186c0a.

[Sharma et al., 2023] Sharma, P. K., Gosain, D., and Diaz, C. (2023). On the anonymity of peer-to-peer network anonymity schemes used by cryptocurrencies. In *Network and Distributed System Security Symposium (NDSS)*. arXiv:2201.11860.

[Shi et al., 2025] Shi, R., Peng, Z., Lan, L., Ge, Y., Liu, P., Wang, Q., and Wang, J. (2025). Eclipse attacks on Monero’s peer-to-peer network. In *Network and Distributed System Security Symposium (NDSS)*.

[The Tor Project, 2026] The Tor Project (2026). Onionoo: details of running relays. Document published 2026-10-05 19:00 UTC, 9,093 running relays. Accessed 2026-10-05.

[Tran et al., 2020] Tran, M., Choi, I., Moon, G. J., Vu, A. V., and Kang, M. S. (2020). A stealthier partitioning attack against Bitcoin peer-to-peer network. In *2020 IEEE Symposium on Security and Privacy (SP)*, pages 894–909.

[X4BNet, 2026] X4BNet (2026). lists_vpn: input/datacenter/ASN.txt. commit 4b2f10491121, 2026-10-05.

A Equivalence of the deduplicated draw

Let G be the set of groups that are not already connected and n_g the number of candidates in group $g \in G$. After a uniform shuffle, the first candidate of group g is each of its n_g members with probability $1/n_g$. After deduplication there are $|G|$ candidates, one per group, and a uniform pick selects each with probability $1/|G|$. A given candidate i in group g is therefore selected with probability $1/(|G|n_g)$, which is the probability of choosing g uniformly from G and then i uniformly from g . The probability that the draw selects an adversary node is $\frac{1}{|G|} \sum_{g \in G} s_g$, where s_g is the share of adversary nodes in group g ; it depends on the number of groups, not on their sizes.

Numerically, with three random peers already connected and 15,000 draws for each of three random seeds, the literal draw of `gen.network()` and the group-then-member draw gave adversary shares within two standard errors of this analytical value for both datasets and both groupings (replication code: `check.equivalence()` in `sim.R`).

B Adversary nodes in groups with honest nodes

Let $S = \sum_{g \in G} s_g$, so that the probability that a draw selects an adversary node is $S/|G|$ (Appendix A). An adversary node placed in a group with $h \geq 1$ honest nodes and x adversary nodes raises s_g by $\delta = h/((x+h)(x+h+1)) \leq 1/2$ and leaves $|G|$ unchanged. The same node in a new group raises S by one and $|G|$ by one. The new group is at least as good for the adversary if $(S+1)/(|G|+1) \geq (S+\delta)/|G|$, that is if $|G|(1-\delta) \geq S+\delta$, which holds for every $\delta \leq 1/2$ when $S/|G| \leq 1/2 - 1/(2|G|)$. While the adversary obtains less than about half of the draws and every group costs the same, it therefore does best by placing nodes in new groups, and the payoffs (1)–(2) are the relevant ones.

C Network centrality

The centrality statistics of the simulated networks, as reported in [Rucknium, 2025a], are given below.

Table 14: Centrality statistics of the network

Scenario	Betweenness	Closeness	Degree	Eigenvector
Oct 2026, /24 dedup, 80% unreachable	1.18×10^{-3}	9.38×10^{-2}	2.50×10^{-3}	7.32×10^{-1}
Oct 2026, AS dedup, 80% unreachable	4.26×10^{-3}	1.05×10^{-1}	1.11×10^{-2}	8.73×10^{-1}
Oct 2026, hybrid, 80% unreachable	2.04×10^{-3}	8.16×10^{-2}	4.51×10^{-3}	8.06×10^{-1}
Oct 2026, /24 dedup, 90% unreachable	1.09×10^{-3}	1.04×10^{-1}	2.34×10^{-3}	7.93×10^{-1}
Oct 2026, AS dedup, 90% unreachable	4.74×10^{-3}	1.08×10^{-1}	1.09×10^{-2}	9.07×10^{-1}
Oct 2026, hybrid, 90% unreachable	2.07×10^{-3}	8.09×10^{-2}	4.40×10^{-3}	8.59×10^{-1}
May 2025, /24 dedup, 80% unreachable	1.04×10^{-3}	7.87×10^{-2}	2.41×10^{-3}	7.60×10^{-1}
May 2025, AS dedup, 80% unreachable	3.75×10^{-3}	1.03×10^{-1}	7.86×10^{-3}	8.76×10^{-1}
May 2025, hybrid, 80% unreachable	1.24×10^{-3}	8.21×10^{-2}	2.82×10^{-3}	7.88×10^{-1}
May 2025, /24 dedup, 90% unreachable	1.07×10^{-3}	8.76×10^{-2}	2.34×10^{-3}	8.17×10^{-1}

May 2025, AS dedup, 90% unreachable	3.97×10^{-3}	1.01×10^{-1}	7.89×10^{-3}	9.05×10^{-1}
May 2025, hybrid, 90% unreachable	1.37×10^{-3}	8.92×10^{-2}	2.83×10^{-3}	8.42×10^{-1}

D Survey sample

Table 15: All sampled ASes and their classification

ASN	Network	Seen	bgp.tools	Verdict
AS20956	SSI-AS	bitcoin		U
AS47610	RWTH-AS	bitcoin+tor		N
AS22750	BCSGroup	tor	dsl	N
AS33363	Charter Communications (33	bitcoin+monero	dsl	N
AS10620	Telmex Colombia S.A.	bitcoin	dsl	N
AS7489	HostUS	tor	vpsh	R
AS48239	IT-TV	tor	dsl	N
AS44574	AS44574 Networks	bitcoin		U
AS1547	JSC InterDnestrCom (IDC)	bitcoin	dsl	N
AS7296	Alchemy Communications, In	bitcoin		R
AS8445	Salzburg AG	bitcoin+tor	dsl	N
AS150706	HKZTCL-AS-AP Hong Kong Zhe	bitcoin		U
AS21581	M5 Computer Security / M5H	bitcoin	vpsh	R
AS46455	NetworkLubbock	bitcoin		U
AS21453	Flex	bitcoin	dsl	N
AS198193	Television por Cable Santa	bitcoin		N
AS215467	Skhron	bitcoin	vpsh	R
AS36103	CentraCom	tor		N
AS58212	Dataforest	bitcoin+tor	vpsh	R
AS33915	Vodafone Libertel B.V.	bitcoin+monero+tor	dsl	N
AS51290	HosTeam	tor	vpsh	R
AS137266	CHINATELECOM-HUBEI-WUHAN-5	monero		N
AS213269	OSS	monero	dsl	N
AS31208	MF-CENTER-AS	monero	dsl	N
AS16075	TAZ	tor		N
AS215590	DpkgSoft International Lim	monero		R
AS812	Rogers Cable	bitcoin+monero+tor	dsl+mobile	N
AS216071	VDSINA	bitcoin+monero+tor	vpsh	R
AS9943	KNCTV-AS KangNam CableTV	bitcoin		N
AS30377	MACST-DUB	monero		U
AS33657	Comcast - Baltimore and D.	bitcoin+monero+tor	dsl	N
AS30404	Blue Stream	bitcoin		N
AS22489	DATABANK-CASTLEACCESS	bitcoin		U
AS30983	MARWAN - Moroccan Academic	monero+tor		N
AS20766	Association Gitoyen	tor		U
AS10013	FreeBit Co,Ltd.	bitcoin	dsl	N
AS4685	ASAHI Net	bitcoin	dsl	N

AS202282	Fibrus	bitcoin+tor	dsl	N
AS15493	RUSCOMP-AS Russian company	bitcoin	dsl	N
AS401401	Unredacted Inc (NoiseNet)	tor		U
AS200736	MEDIANET INVEST M.I.K.E	bitcoin+monero+tor	dsl	N
AS10996	VIF Internet	tor		N
AS19949	FNFISP	tor		U
AS24877	UAB INIT Kns	monero		U
AS56030	Voyager Internet	monero+tor	vpsh+dsl	R
AS142108	Rotko Networks	bitcoin		U
AS202053	UpCloud	bitcoin+tor	vpsh	R
AS141356	Universe Action	bitcoin		U
AS202596	G.Network Communications	monero	dsl	N
AS42675	Obehosting	bitcoin+monero+tor	vpsh	R
AS16302	SuomiCommunications	bitcoin+monero+tor	vpsh+dsl	R
AS965	WEBHOSTINGHOLDINGS	tor		U
AS200295	MT-SKOED	bitcoin		U
AS198985	Aquilenet	tor		U
AS12310	iNES Group	monero		R
AS44512	KONVERTO	bitcoin		U
AS40244	Turnkey Internet	tor	vpsh	R
AS56546	KTASTRA-AS	bitcoin		U
AS9158	Telenor Denmark	tor	dsl	N
AS34700	Maxnet Ukraine	bitcoin+monero+tor	dsl	R
AS203953	HIPER	bitcoin+tor	dsl	N
AS49531	NETCOM-R-AS	tor		U
AS25513	MGTS	bitcoin	dsl	N
AS29909	Metro Optic	bitcoin		U
AS18106	ViewQwest	tor	dsl	N
AS19799	HCS	tor		U
AS5408	National Infrastructures f	tor		N
AS54936	Wyverd Group	bitcoin+tor		N
AS1448	United Cooperative Service	bitcoin		N
AS57794	HCN	bitcoin	dsl	N
AS4364	IGLOU	bitcoin		U
AS35366	ISPpro Internet KG	tor	vpsh	R
AS5607	Sky Broadband	bitcoin+tor	dsl+mobile	N
AS4764	Aussie Broadband	bitcoin+monero+tor	dsl	N
AS208208	CYBERSE	bitcoin+tor	vpsh	R
AS34941	CYBERCOM-AS	bitcoin		U
AS396948	CLOUDWEBMANAGE-SC	monero		U
AS6057	ANTEL	bitcoin	dsl	N
AS270039	PRILZ COLOMBIA	tor		U
AS61900	Mais Internet	bitcoin		N
AS12668	MiraLogic Telecommunicatio	bitcoin+monero	dsl	N
AS51561	ICUK Computing Services Lt	bitcoin	dsl	N
AS200303	LUMASERV	tor	vpsh	R
AS42532	Veesp SIA	tor	vpsh	R

AS200950	Calibour Limited	tor	vpsh	R
AS45899	VNPT	bitcoin+monero+tor	vpsh+dsl+mobile	R
AS5617	Orange Polska - TPNET	bitcoin+monero+tor	dsl+mobile	N
AS12430	Vodafone España	bitcoin+tor	dsl+mobile	N
AS197648	CL8ASN1	tor		U
AS47	University of Southern Cal	bitcoin		N
AS206264	Amarutu Technology Ltd.	tor	vpsh	R
AS204272	ASARTA	bitcoin		U
AS396325	The Fusion Networks	bitcoin		N
AS399122	VPS House Technology Group	tor	vpsh	R
AS49791	3H CLOUD	bitcoin	vpsh	R
AS24233	SUPERLOOP-AS-AP SUPERLOOP	bitcoin		N
AS25353	BAR Informatik Net	tor		U
AS50771	GIGALINK-LTD-AS	bitcoin	dsl	N
AS395502	Sunbreak Electronics	bitcoin+monero		U
AS399512	GREATWAVE-HE-A	bitcoin		U
